

JAMES MATTHEW HARRISON

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Location: Birmingham, United Kingdom

PROFESSIONAL SUMMARY

I'm a Data Engineer with six years of hands-on experience building and maintaining data pipelines that process billions of records daily. I've spent the last three years at mid-sized tech companies where I've led infrastructure decisions and mentored junior engineers through complex ETL challenges. My background is in writing clean, testable code and understanding how data flows through systems from ingestion to analytics. I care about operational excellence and writing documentation that actually helps people understand what's happening.

WORK EXPERIENCE

Senior Data Engineer

DataFlow Systems

dataflowsystems.io

Birmingham, United Kingdom

March 2022 - Present

Designed and implemented a real-time data platform processing 500 million events daily using Apache Kafka and Spark, reducing query latency by 65% for downstream analytics teams.

Migrated legacy batch processes from Talend to Python-based Airflow DAGs, cutting infrastructure costs by 40% while improving pipeline visibility and error handling.

Established data quality frameworks using Great Expectations, catching data anomalies upstream and preventing 12 major incidents from reaching production over 18 months.

Mentored four junior engineers on SQL optimization, cloud architecture, and debugging distributed systems, with two promoted to mid-level roles under my guidance.

Data Engineer

CloudMetrics Ltd

cloudmetrics.co

Birmingham, United Kingdom

June 2020 - February 2022

Built and maintained 80+ production data pipelines in AWS using PySpark and Python, supporting 150+ data analysts and machine learning engineers across the company.

Optimized Redshift queries reducing average query execution time from 45 seconds to 8 seconds, directly improving user experience for BI tools serving 300+ stakeholders.

Implemented automated data validation and monitoring using Python and SQL, creating alerting systems that reduced time to detect failures from hours to minutes.

Collaborated with analytics and product teams to design new data schemas and dimensional models supporting A/B testing and customer segmentation initiatives.

Junior Data Engineer

TechVenture Solutions

techventuresolutions.net

Manchester, United Kingdom

September 2018 - May 2020

Developed ETL processes using Python and SQL to extract data from 15+ source systems into a centralized data warehouse, supporting company-wide reporting needs.

Automated manual data processes using shell scripts and Python, reducing team workload by approximately 20 hours per week and minimizing human error in critical reports.

Learned cloud fundamentals and contributed to the company's initial migration from on-premise databases to AWS RDS and S3 infrastructure.

Documented data lineage and created runbooks for common troubleshooting scenarios, helping support team resolve 85% of data issues without escalation to engineering.

EDUCATION

MSc Data Science

University of Birmingham

Birmingham, United Kingdom

Graduated September 2018

BSc Computer Science

Aston University

Birmingham, United Kingdom

Graduated June 2016

TECHNICAL SKILLS

Languages: Python, SQL, Scala, Bash

Data Technologies: Apache Spark, Apache Kafka, Airflow, Talend

Cloud Platforms: AWS (S3, RDS, Redshift, EC2, Lambda), Google Cloud Platform

Databases: PostgreSQL, MySQL, Snowflake, Redshift, DynamoDB

Tools and Frameworks: Pandas, NumPy, Jupyter Notebooks, Great Expectations, dbt

Version Control: Git, GitHub, GitLab

Other: Linux, Docker, JIRA, Confluence, Data Modeling, ETL Design